be able to realize the gains from trade that are available in this situation and are not currently occurring because of the lack of information. Market participants that post the odd-lot quotes that these market participants trade against would also benefit from realizing additional gains from trade.

The magnitude of this benefit depends on the amount of additional or improved trading generated by the inclusion of odd-lot information. In particular, the Commission believes that to the extent market participants who rely solely on SIP data and lack information on odd-lot quotes choose to receive the odd-lot information in expanded core data and would have traded frequently against odd-lot quotes had they known about them, the benefit will be large. However, if these market participants would not have frequently traded against odd-lot quotes but for a lack information, then the Commission believes that the associated economic benefit from including odd-lot quotes in core data will be small. The Commission believes it is not possible to observe this willingness to trade with existing market data.

Market participants who choose to receive the odd-lot quotes, or their clients, may benefit from making more informed trading decisions by utilizing the data to improve their strategies related to order routing and order placement, provided that they do not already obtain information on odd-lots from proprietary feeds. For instance, market participants who wish to fill an order at the best possible price, including at sizes of less than 100 shares, will be better able to do so because odd-lot quotes at prices better than the NBBO will be visible to them. Additionally, these market participants may be able to improve the placement of their limit orders by being able to see odd-lot quotes at or inside the NBBO at multiple exchanges in order to evaluate which exchange's queue would provide their limit order with the highest execution priority. The use of this information may improve order execution quality and facilitate best

execution for these market participants or their clients.[1962] The Commission believes that many of the market participants who utilize such strategies already have access to full odd-lot information via proprietary feeds; for these market participants, this portion of the final amendments may not improve their strategies related to order routing.[1963]

Also, the Commission believes that some market participants might start running these order routing strategies if the data were available to them at prices that are lower than the cost of obtaining this data through proprietary feeds.[1964] These market participants might currently find that the value of attempting such strategies without information on odd-lots is too low to justify running the strategies, but they might find that access to data on such orders through the updates to expanded core data will enable them to run such strategies effectively. To the extent that such market participants exist, the inclusion of odd-lot quotes in core data will be a benefit to them as well.[1965]

1962 For a discussion of order execution quality and the provision of execution services by broker-dealers, see supra Section V.B.3(e).

1963 Adding information on odd-lot quotes priced at or better than the NBBO to expanded core data may benefit those market participants who already obtain odd-lot information by providing them with alternatives to proprietary feeds. For a discussion of this effect, see infra Section V.C.4(a). Also, the Commission understands that some market participants who use proprietary feeds as their main source of market data also use the SIP feeds as a backup. For such market participants, adding information on odd-lot quotes priced at or better than the NBBO to expanded core data may improve the value of a core data feed as a backup if they choose to subscribe to the additional information contained in expanded core data.

1964 See infra Section V.C.2(b).

1965 For further discussion of new entrants to the competitive order routing business, see infra Section V.C.4(b).

The Commission believes that adding information on odd-lot quotes priced at or better than the NBBO to expanded core data may improve price efficiency. The wider availability of information about odd-lot quotes may mean that market participants who currently do not receive this information and subscribe to this element of expanded core data will incorporate the information contained in those quotes into their trading decisions. This may have the effect of improving the efficiency with which this information becomes reflected in prices.[1966]

One commenter stated that adding information on unprotected odd-lot quotations to core data would create confusion for retail investors.[1967] The Commission disagrees with this commenter. As discussed above in this section, the Commission believes that many retail brokers will not directly offer their clients all of the odd-lot information contained in expanded core data and, therefore, their clients will not be confused by it. If a retail broker does directly offer all of the information to any of its clients, the Commission believes that any client receiving the information will likely be a sophisticated retail investor and not confused. Additionally, if competing consolidators develop products for retail brokers to offer to their clients (i.e., retail investors) that contain subsets of the odd-lot information in expanded core data, the Commission believes that competing consolidators and the data vendors or broker-dealers that supply the information to retail investors will do so in way that does not create confusion.

The Commission believes that adding information on odd-lot quotes priced at or better than the NBBO to expanded core data may cause changes to order flow as market participants that do not currently receive this information and choose to subscribe to it change their trading

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1966 For additional discussion of the price efficiency point, see infra Section V.D.1.

1967 See TD Ameritrade Letter at 4–5.

strategies to take advantage of newly visible quotes. This may mean that there will be changes to the share of order flow each exchange and off-exchange trading center receives as a result of this rule. The Commission is uncertain about the magnitude of this effect.

The addition of odd-lot quote information to expanded core data will increase the total message traffic in expanded core data, and this increase in message traffic will be accompanied by costs to market participants to set up the infrastructure required to handle this new level of traffic. Additionally, competing consolidators and SROs may incur implementation costs related to receiving and generating the information necessary to process and disseminate consolidated market data. However, market participants are not required to receive (or display) the odd-lot quotes contained in expanded core data, and competing consolidators will not be required to disseminate all of the information in consolidated market data, including odd-lot quotes contained in expanded core data, so they will not incur these costs unless they choose to receive or disseminate this information, respectively.[1968] These costs are discussed below in Section V.C.1(c)(iv).

The addition of information on all odd-lot quotes priced at or better than the NBBO to core data may negatively affect certain trading strategies, but the associated costs are likely to be small. First, the Commission believes that there may be traders who currently attempt not to display their orders to wide public view by posting them in odd-lot sizes, in pursuit of trading strategies that take advantage of a market's limited knowledge of odd-lot size quotes. The Commission understands that certain traders (ones who are the most likely to recognize any advantage being sought in this manner) obtain proprietary feeds and so currently can see these odd-lot quotes. This means that this strategy cannot be used to hide quotes from users of

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[1968] See supra Sections II.C.2(a) and III.C.8(a)(ii).

proprietary DOB feeds. To the extent that it is necessary to hide the quotes from such users in order for the strategy to work, the benefits of such a trading strategy are likely to be minimal. If this is the case, then to the extent that the addition of odd-lot quotes to core data makes this strategy more difficult, the Commission believes that the cost to these traders of losing such an opportunity will also be minimal. On the other hand, if there is some benefit to posting quotes in odd-lot sizes to hide them from view (or at least from the view of market participants that do not observe these odd-lot quotes) despite the fact that users of proprietary DOB feeds can still see the quotes, the Commission believes that to the extent that the addition of odd-lot quotes to core data makes this strategy more difficult, there may be a cost to the traders who use such a strategy. The Commission cannot observe whether an odd-lot quote is being used to hide the order or not.

Second, there may be costs to those traders who currently enjoy the position of being among the traders who can see odd-lot quotes via proprietary data feeds. The Commission believes that odd-lot quotes are more easily taken advantage of by those traders who can see the quotes. Currently, this advantage is available only to those traders who purchase proprietary data feeds. The Commission believes that this gives these traders an advantage over other traders by improving their order execution costs. Under the changes to core data, this advantage is likely to be reduced. If this were to happen, it will be because other traders will obtain the advantage as well and may take advantage of these quotes before the current direct feed subscribers do. To the extent that this happens, this cost to current direct feed subscribers from losing this advantage represents a transfer to the traders who can see the liquidity currently in odd-lots. The Commission is uncertain about the size of the loss in advantageous trading opportunities to traders who subscribe to the proprietary data. To quantify this requires knowing (among other things) when an odd-lot quote is traded with by a participant who had access to full odd-lot

information and when it was traded with by a participant who did not know the quote was there, and this is not observable from available market data.

It can sometimes happen that a market becomes locked or crossed in odd-lot orders. As a result of the final amendments, information on all odd-lot quotes priced at or better than the NBBO will now be included in expanded core data, and these locked and crossed odd-lot orders will now be visible to subscribers of expanded core data that chose to receive odd-lot information. The economic effects of having these locked or crossed quotes visible to market participants who receive this data will be minor. In particular, to the extent that these crosses and locks in odd-lot sizes represent a profitable trading opportunity to those market participants who do not receive odd-lot information, being able to observe the occurrence of these events as a result of the receiving odd-lot quotes in expanded core data will be a benefit to these market participants. Also, to the extent that market participants who currently subscribe to proprietary feeds are able to profit from being the only market participants to observe crossed or locked odd-lots, the change will represent a cost to them. To the extent these market participants can profit from exploiting those market participants who cannot see the crosses or locks, this change will represent a transfer from those who currently trade on this information to those who acquire the information through new core data and are able to use it effectively. It is also possible that traders avoid sending orders because of the risk of being exploited if they cross or lock the market. To the extent that this happens and that the expansion of core data addresses this concern, the increase in trading that will result will represent a benefit to both sides of the trade. The Commission believes that some crossed or locked odd-lot quotes represent traders who are not aware at the time they post their quote that the quote could be filled by a marketable order elsewhere. To the extent this happens it represents a cost to this trader since the posted order is

exposed to the risk that it will be executed with a marketable order at a price inferior to what is available on the market to the trader who posted the order. The final amendments will reduce this cost for market participants who receive odd-lot information because they will now be able to observe and trade with odd-lot orders available at better prices.

(ii) Effects of Addition of Depth of Book Information

The Commission is adding certain depth of book information to the definition of core data, which will result in this information becoming available to anyone who subscribes to this element of core data. The Commission believes that this information could be useful in trading, and therefore disseminating this information as an element of core data could have the effect of causing changes to the trading strategies of those market participants who currently rely solely on SIP data and will choose to buy depth of book information. This could potentially lead to improvements in order routing for these market participants or their clients and reductions in their execution costs and facilitate best execution. Adding certain depth of book information to the definition of core data may also lead to changes in order flow to trading venues, improvements in price efficiency of markets, and gains from trade that are not currently being realized. Market participants that choose to receive the depth of book data may experience implementation costs from having to upgrade infrastructure to account for the increase in message traffic from the data.

Some commenters stated that most market participants do not need depth of book information.[1969] However, other commenters believed that including depth of book data in core

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[1969] See, e.g., Nasdaq Letter IV at 33; TD Ameritrade Letter at 5.

data would be useful for market participants.[1970] The Commission recognizes that many market participants, including many retail investors, may choose not to receive all of the DOB information contained in expanded core data.[1971] However, the Commission believes that there are some market participants that currently do not receive DOB information but may choose to receive this information from expanded core data if it is available at a lower price than equivalent proprietary DOB feeds.[1972] If these market participants subscribe to this element of core data, then the Commission believes they will receive many of the benefits (and incur many of the costs) discussed below. Even if market participants do not directly receive all of the DOB information in expanded core data, they could realize some of the benefits if competing consolidators offer products that are derived from or contain some of the DOB information in expanded core data. For example, competing consolidators could offer a product that contains only information on the price and size available at the next best round lot price outside the NBBO. Because such a product would not significantly increase message traffic compared to receiving all DOB information in expanded core data, many market participants, including many retail brokers (who may offer it to their clients), may be able to utilize such a product and gain additional information that would allow them to lower their execution costs. Even if market participants do not receive any additional DOB information contained in expanded core data, they may still benefit indirectly from including depth of book information in core data if the broker-dealers handling their orders use the information. If a broker-dealer previously did not

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1970 See, e.g., Clearpool Letter at 14; Healthy Markets Letter I at 3; DOJ Letter at 2–4.

1971 See supra Section V.C.1(c).

1972 See infra Section V.C.2(b).

have access to DOB information, then its clients may benefit if a broker-dealer uses the DOB information in expanded core data when handling customer orders, which may improve their execution quality.[1973] Additionally, the Commission believes that the depth of book information in expanded core data may benefit market participants who substitute it for proprietary DOB feeds if it is available at lower cost [1974] The Commission is not able to quantify the number of market participants who will directly utilize the depth of book information in core data because it would depend on the future fees the Equity Market Data Plan establishes for the additional content of core data.

One commenter stated that it was unclear if five levels of depth would be useful to any investors at all.[1975] However, the Commission believes, and commenters agree, that including five levels of depth in expanded core data will benefit market participants.[1976] The Commission acknowledges that market participants that substitute expanded core data for proprietary DOB feeds will not receive as much depth of book information and may experience a reduction from the benefits they receive from such information, However the Commission believes that these market participants will only substitute expanded core data for proprietary DOB data if the

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1973 See supra Section V.B.2(a). Even if a broker-dealer previously received the data from proprietary feeds and now receives it from core data, clients of the broker-dealer may benefit if the broker-dealer indirectly passes on any cost savings from switching data to its clients.

1974 See infra Sections V.C.2(b)(i) and V.C.4(a).

1975 See Nasdaq Letter IV at 33.

1976 Commenters agreed that five levels of depth is sufficient for many market participants. See, e.g., State Street Letter at 2–3; Capital Group Letter at 3; Fidelity at 4. In addition, the staff analysis found a significant percentage of the total notional value of all depth of book quotations for both liquid and illiquid stocks falls within the first five price levels. See supra note 387. See also supra Section II.F.2(b).

money they save exceeds the value of the reduction in benefits from not receiving the additional information contained in proprietary DOB feeds.

The Commission believes that adding the depth of book information as an element of core data will benefit market participants who previously relied exclusively on SIP data and who choose to receive this element of expanded core data. Academic research has found evidence that valuable trading information can be obtained from the full depth of a limit order book.[1977] As noted in the Proposing Release, some market participants also believe that depth of book information is valuable.[1978] Currently, only traders who subscribe to exchanges' proprietary data feeds can receive this information. As a result of the final amendments, additional depth of book information will become available to anyone who subscribes to these elements of core data. The Commission believes that market participants, including, as suggested by commenters, some retail investors,[1979] that currently rely solely on SIP data could use the additional depth of book

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1977 See Lawrence E. Harris and Venkatesh Panchapagesan, The Information Content of the Limit Order Book: Evidence from NYSE Specialist Trading Decisions, 8 J. FIN. MKTS. 25 (2005); Jonathan Brogaard et al., Price Discovery without Trading: Evidence from Limit Orders, 74 J. FIN. 1621–58 (2019); Shmuel Baruch, Who Benefits from an Open Limit-Order Book?, 78 J. BUS 1267 (2005) (presenting some theoretical results showing that liquidity takers benefit more from an open limit order book).

1978 See Proposing Release, 85 FR at Section III.C.2(c) (describing how market participants have stated that they believe they need depth of book information in order to run their businesses). See also supra Section V.B.2(a) (discussing the value of depth of book information).

1979 See, e.g., Schwab Letter at 1, 3 ("providing depth-of-book data on the consolidated feed will give Main Street investors a critical look at market sentiment with regard to an individual security and pricing information for the size of the order they want to place"); Angel Letter at 8.